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DURING A MEASUREMENT, INFORMATION IS **EXTRACTED** FROM A QUANTUM SYSTEM, AND IN THIS PROCESS THE STATE CHANGES

1. THE STATE IS PROJECTED ONTO ANOTHER STATE (PROJECTION RULE)



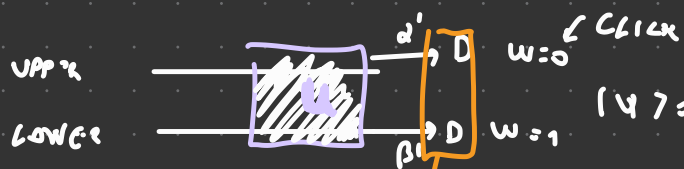
$$P_z = |\langle z | \psi \rangle|^2$$

2. THE STATE IS MODIFIED IN ANOTHER WAY

$$D \sim D_0$$

$$D \sim D_1$$

IN THE MEASUREMENT THE PHOTON IS DESTROYED



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\alpha|^2 + |\beta|^2 = 1$$

$$\begin{cases} \langle w | 1 \rangle = 0 \\ \langle w | w \rangle = \langle 1 | 1 \rangle = 1 \end{cases}$$

THE DETECTION IS NOT UNITARY,

$$P_w = P_d = 0$$

$$\mathcal{H} = \text{SPAN}\{|w\rangle, |d\rangle\}$$

$$\mathcal{H} = \text{SPAN}\{|w\rangle, |d\rangle, |0\rangle\}$$

↓ ZERO
NO PHOTONS

Before $|\psi\rangle \rightarrow$ AFTER THE MEASUREMENT $|\phi\rangle$



$$\mathcal{H} = \text{SPAN}\{|w\rangle, |d\rangle\}$$

$$P_w(\phi) = \begin{bmatrix} e^{i\phi} & 0 \\ 0 & 1 \end{bmatrix}$$

$$P_w(\phi)|w\rangle = e^{i\phi}|w\rangle$$

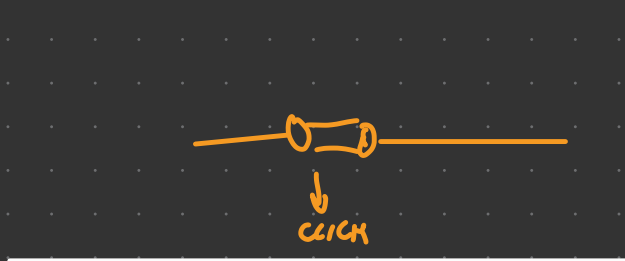
$$P_w(\phi)|d\rangle = |d\rangle$$

$$\mathcal{H}' = \text{SPAN}\{|w\rangle, |d\rangle, |0\rangle\}$$

$$P_{w'}(\phi) = \begin{bmatrix} e^{i\phi} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P_{w'}(\phi)|\phi\rangle = |\phi\rangle$$

$$\mathbf{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \mathbf{X}' = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



EVOLUTION THE STATE OF A QUANTUM SYSTEM MAY ALSO CHANGE OVER TIME EVEN WHEN THE SYSTEM ISN'T BEING MEASURED

$$\begin{array}{l} E_1 \text{ --- } |E_1\rangle \\ E_0 \text{ --- } |E_0\rangle \end{array}$$

$$|\psi(t)\rangle = e^{-i\frac{E_n}{\hbar}t} |E_n\rangle$$

$$\bullet |\psi\rangle \rightarrow |\psi'\rangle$$

$$\bullet |\phi\rangle \rightarrow |\phi'\rangle$$

$$a|\psi\rangle + b|\phi\rangle \rightarrow a|\psi'\rangle + b|\phi'\rangle$$

EVOLUTION RESPECT THE PRINCIPLE OF SUPERPOSITION AND IS A LINEAR MAP IN THE HILBERT SPACE $\mathcal{H}$

$$|\psi(t_2)\rangle = U(t_2, t_1) |\psi(t_1)\rangle$$

THE EVOLUTION OPERATOR $U(t_2, t_1)$ HAS A NUMBER OF PROPERTIES

$$\bullet \text{ IT IS UNITARY: } U^\dagger U = U U^\dagger = \mathbb{I}$$

• THE EVOLUTION OPERATOR PRESERVE THE INNER PRODUCT

$$|\psi\rangle = U|\psi\rangle \quad |\phi\rangle = U|\phi\rangle \Rightarrow \langle U|\psi\rangle, U|\phi\rangle = \langle \psi|U^\dagger U|\phi\rangle$$

$$U(t_3, t_1) = U(t_3, t_2) U(t_2, t_1)$$

U IS NORMAL OPERATOR

ANY UNITARY OPERATOR IS A NORMAL OPERATOR

$$U|n\rangle = e^{i\alpha_n} |n\rangle$$

$$\alpha_n \in \mathbb{R}$$

$$U = \sum_n e^{i\alpha_n} |n\rangle \langle n|$$

PHYSICALLY THE UNITARY TIME EVOLUTION OF A SYSTEM IS DETERMINED BY THE **INTERNAL** PROPERTIES OF THE SYSTEM AND VARIOUS EXTERNAL PARAMETERS



A PARTICULAR CLASS OF UNITARY EVOLUTION IS THAT OF UNIFORM DYNAMICS IN WHICH THE UNITARY OPERATOR DEPENDS ONLY ON INITIAL AND FINAL TIMES

FOR UNIFORM DYNAMICS WE HAVE $U(E_2, E_1) = U(E_2 - E_1)$

IT IS USEFUL TO DEFINE THE CLASS OF OPERATORS $U(E)$ WITH $U(0) = \mathbb{I}$

THESE OPERATORS HAVE SEVERAL PROPERTIES

$$[U(E), U(S)] = 0 \quad U(E)U(S) = U(E+S) \quad \begin{array}{l} \uparrow \\ \text{UNIFORM DYNAMICS} \\ \downarrow \\ U(S)U(E) = U(E+S) \end{array} \Rightarrow \boxed{U(E)U(S) - U(S)U(E) = 0}$$

$$\begin{cases} U(S)|k\rangle = e^{i\alpha(S)}|k\rangle \\ U(E)|k\rangle = e^{i\alpha(E)}|k\rangle \end{cases}$$

$$\begin{aligned} U(S)U(E)|k\rangle &= U(S)e^{i\alpha(E)}|k\rangle = e^{i[\alpha(S) + \alpha(E)]}|k\rangle \\ U(S+E)|k\rangle &= e^{i\alpha(S+E)}|k\rangle \end{aligned}$$

$$\alpha(S+E) = \alpha(S) + \alpha(E)$$

$\alpha(x)$ MUST BE LINEAR

$$\alpha'(0) = 0 \Rightarrow \alpha(x) = \alpha x$$

$$U(x)$$

SO WE CAN WRITE $U(E) = \sum_k e^{-i\omega_k E} |k\rangle\langle k|$

$$U(t) = \sum_k e^{-i\omega_k t} |k\rangle \langle k| \quad \text{6.6. Defining}$$

$$V(t) |1\rangle = e^{i\frac{\Omega}{2}t} |2\rangle$$

$$U(t) |2\rangle = e^{i\frac{\Omega}{2}t} |2\rangle$$

$$U(t) = e^{-i\lambda}$$